Noel Andolz Aguado

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Portfolio Website

EDUCATION

Carlos III University of Madrid

Telecommunications Engineering & Cybersecurity Dual Master — 2025–2027

Bachelor's Thesis - "Measurement and Characterization of Radio Signals in Amplitude and Phase" — 2024–2025

Telecommunication Technologies Engineering at San Francisco State University — 2022–2023

Academic exchange year at San Francisco State University within the Universidad Carlos III de Madrid Non-European Mobility program.

Bachelor's Degree in Telecommunication Technologies Engineering — 2020–2025

PROFESSIONAL EXPERIENCE

Internship | Ericsson — **April 2025 – May 2026**

- Contributing to the testing team for Ericsson's Cloud Native 5G solutions, with a focus on enhancing the efficiency of the system testing phase through data modelling and artificial intelligence techniques. The work involves orchestrating containerised test environments with **Kubernetes**, and leveraging AI-assisted development tools including **Amazon Q** and **Kiro** to accelerate test automation workflows and improve overall code quality.

Internship | Accenture — **October 2024 – February 2025**

- As an intern in the Tech Advance Testing team, I contributed to creating and implementing comprehensive testing strategies, gaining hands-on experience in quality assurance processes within a large-scale enterprise environment.

Area VIP | Real Madrid — **September 2023 – August 2025**

- As a VIP Area Attendant at Santiago Bernabéu Stadium, responsible for ensuring distinguished guests experience the full extent of our premium services, managing high-pressure situations with professionalism and attention to detail.

Waiter and Barman | La Ciboulette — **June 2020 – July 2024**

- Developed practical skills in teamwork, serving procedures, and event organisation at Ciboulette Catering. This role reinforced the value of collaboration in fast-paced, client-facing environments.

PROJECTS

Personal Portfolio Website — **2026 (view project)**

- Bilingual responsive portfolio website built with vanilla HTML5, CSS3, and JavaScript, featuring a custom translation system with localStorage persistence for language preference retention across sessions. Implements a mobile-first responsive design across three breakpoints (968px, 640px, 375px) with touch-optimized UI components (44×44px minimum touch targets per WCAG guidelines), hamburger menu with overlay navigation, and smooth scroll with active section highlighting. Showcases 14 technical projects spanning cloud-native architectures, embedded systems, machine learning, and SDN/network programming, organized into four skill categories: Programming Languages, Cloud Native & DevOps, Networking & Telecommunications, and Frameworks & Tools.

Modular Pentesting Automation Framework — **2026**

- Modular security auditing framework developed in Python and Robot Framework during the Ericsson internship, designed to automate the scanning and analysis of SSL/TLS and SSH services across cloud-native infrastructure. Features an interactive CLI that orchestrates independent test modules through a three-layer architecture (menu, Robot orchestration, Python libraries), covering cipher suite classification against a database of 50+ algorithms, protocol version detection, and automated pattern search across PDF documentation archives for exposed credentials, tokens, and keys using 18 regex-based detection rules. Generates per-scan text reports with per-finding severity ratings (BEST/GOOD/ACCEPTABLE/WEAK/INSECURE) and Word documents summarizing pattern matches across hundreds of extracted PDFs.

Cloud-Native Testing Visualization Platform (TCDB) **2025**

- Development of TCDB (Test Case Database), a cloud-native platform built during the Ericsson internship within the BCSS CNE PDU UDM Solutions team to centralise and visualise Jenkins CI/CD test results across 5G functional test pipelines. The system follows a microservices architecture deployed on Kubernetes via Helm charts, comprising a React frontend, a Node.js backend exposing RESTful APIs, a FastAPI + SQLAlchemy + Pydantic ingestion API that parses Jenkins XML output into a MySQL database, and Grafana dashboards tracking pass rates, failure trends, and execution statistics over rolling 24-hour, 7-day, and 30-day windows. The solution adopts a multi-tenant design, enabling independent teams to share a common infrastructure while maintaining team-specific views and endpoints.

Mininet and Ryu **2025 (view project)**

- Development of Software-Defined Networking (SDN) scenarios using Mininet and the Ryu controller. The project involved emulation of network topologies, configuration of OpenFlow-based switches, and implementation of custom control logic in Python, including MAC learning, flow table management, and packet forwarding strategies. Additional work covered automation of network experiments through the Mininet Python API, integration with remote controllers, and analysis of ARP and OpenFlow protocol behaviour in controlled virtual environments.

SDN Development Lab **2025 (view project)**

- Design and implementation of an IPv4 software router using Ryu within a Mininet-based emulated environment. Custom OpenFlow control logic performs Layer 3 packet forwarding including routing table matching, TTL manipulation, and dynamic header rewriting. Extended to support ICMP echo reply generation and dynamic ARP resolution, enabling full end-to-end connectivity validated through traffic analysis and packet inspection.

IP Route Lookup using Compressed Trie **2023 (view project)**

- An IP route lookup implementation using the compressed trie (Patricia trie) algorithm, efficiently searching for the longest prefix match in a routing table for given IP addresses.

Participation in Vodafone Challenge **2023**

- Participated in the 2023 Vodafone Challenge as part of a Mobile Communications course. Attended Vodafone's Madrid headquarters for sessions on 5G, Massive MIMO, RAN and Open RAN, then led a group project on Low Layer Split options in Downlink, presenting results in front of senior Vodafone engineers. A hands-on experience emphasising technical depth and cross-functional teamwork.

Dijkstra Algorithm **2022 (view project)**

- Implementation of Dijkstra's Algorithm to compute shortest paths, developed for the ENGR 476 Computer Communications Networks course at SFSU. Uses a cost matrix as input where non-connected nodes are represented with a finite cost value.

STM32 Reaction Games **2021 (view project)**

- Firmware for two two-player reaction games on the NUCLEO-L152RE board, controlled through buttons, LEDs, a buzzer, and a potentiometer. Progressively integrates GPIO configuration, hardware timer interrupts (TIM2/TIM4), ADC-based potentiometer reading, PWM-driven buzzer melodies, and UART serial communication.

Process Scheduler Emulator **2021 (view project)**

- A C-based application emulating the simplified behaviour of a process scheduler, developed as part of the Systems Architecture course at Carlos III University of Madrid.

ADDITIONAL INFORMATION

International Exchange:

- San Francisco, California, USA 2022
- Maryland, USA 2019
- Totnes, England 2017

Other Studies:

- Music studies at Escuela-Conservatorio Manuel Rodríguez Sales.
- Videogame programming course hosted by Comunidad de Madrid.
- Robotics course hosted by IES Isaac Albéniz.

IT Skills:

C, Java, Python, MATLAB, VHDL, Kubernetes, Git/GitHub, ANSYS HFSS, SQL, LTSpice, Excel, AWR

Volunteering:

- Red Cross PINEO and OTL programs 2019–2020
- Participation in Scouts groups 101 and 601

LANGUAGES

English: Advanced professional proficiency
Spanish: Native
German: Basic

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